



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICRO SERVICES (24SDCS03)

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TITLE OF THE PROJECT	HealthVerse AI: An Agentic Digital Healthcare Ecosystem for Predictive Health Monitoring and Personalized Wellness	
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Abstract:

HealthVerse AI: An Agentic Digital Healthcare Ecosystem for Predictive Health Monitoring and Personalized Wellness proposes an intelligent and integrated digital healthcare ecosystem that aims to transform conventional healthcare monitoring into a more proactive, predictive, and personalized approach. Existing digital healthcare solutions often provide individual services such as fitness tracking, symptom monitoring, or basic health assistance, but they may not provide a unified mechanism for continuously analyzing diverse health and lifestyle information, identifying emerging health risks, and generating personalized wellness guidance. HealthVerse AI addresses this gap by combining artificial intelligence, predictive health monitoring, personalized wellness, and agentic AI within a single digital healthcare framework.

The proposed ecosystem is designed to collect and process relevant health and lifestyle information from available sources. Such information may include parameters related to physical activity, sleep patterns, heart rate, blood pressure, blood glucose, body measurements, lifestyle habits, and other relevant wellness indicators. The collected information is processed to identify meaningful patterns, trends, and changes in an individual's health profile. By applying appropriate artificial intelligence and

machine learning techniques, the system aims to support predictive health monitoring by identifying potential risk patterns at an earlier stage rather than relying only on reactive responses after a health issue has occurred.

A major aspect of HealthVerse AI is the incorporation of an agentic AI architecture. Instead of depending on a single AI component for all healthcare-related tasks, the proposed ecosystem can utilize specialized intelligent agents that perform complementary functions. A Health Monitoring Agent can focus on observing and analyzing health-related parameters and changes over time. A Personalized Wellness Agent can generate recommendations based on an individual's health profile, lifestyle patterns, and wellness objectives. A Health Assistant Agent can provide an interactive interface through which users can obtain relevant information and assistance. The collaboration of these agents is intended to create a continuous cycle of monitoring, analysis, prediction, recommendation, and feedback.

The personalization component is an important part of the proposed ecosystem. Instead of providing identical recommendations to every user, HealthVerse AI is intended to consider individual health information, lifestyle patterns, and personal wellness goals when generating recommendations. This can support areas such as physical activity, sleep management, healthy lifestyle practices, and general wellness awareness. The system can continuously incorporate new information and update the user's health profile, allowing the generated insights and recommendations to become more relevant to the individual's changing health patterns.

The proposed system also focuses on addressing the research gap between health monitoring and intelligent predictive assistance. While AI has demonstrated potential in healthcare-related analysis and prediction, an integrated ecosystem that combines continuous monitoring, predictive analysis, personalized wellness, and multiple collaborating AI agents provides an opportunity for a more comprehensive digital healthcare experience. HealthVerse AI therefore emphasizes not only the analysis of health data but also the intelligent coordination of different healthcare-support tasks through an agentic framework. The system is intended to function as a digital health and wellness decision-support platform and should not be considered a replacement for professional medical diagnosis or treatment.

The expected contribution of HealthVerse AI is the development of a unified and intelligent healthcare ecosystem that brings together predictive health monitoring, personalized wellness support, and agentic artificial intelligence. By integrating these capabilities, the proposed system aims to improve health awareness, support early identification of potential risk patterns, encourage healthier lifestyle decisions, and provide users with continuous and personalized digital assistance. The proposed architecture provides a scalable foundation for integrating additional AI agents, predictive analytics modules, healthcare services, and personalized wellness functionalities, allowing the HealthVerse AI ecosystem to evolve according to future healthcare requirements.

Overall, HealthVerse AI aims to bridge the gap between conventional digital health monitoring and intelligent, proactive healthcare assistance by creating an agentic ecosystem capable of continuously analyzing health information, supporting predictive insights, and delivering personalized wellness guidance.

Project Architecture/Project Overview:

